(a) Methodological Ground Simulation Standards (3-Chamber Parity)

Chamber 1: Geomagnetic Baseline (GMF)

Magnetic Regime: 45–50 μT Ambient Terrestrial Field

Experimental Purpose: Terrestrial environmental control (calibrated local vector)

Chamber 2: Near-Null Field Simulation (NNMF)

Magnetic Regime: <10 nT (Shielded / Actively Cancelled)

Experimental Purpose: Lunar surface magnetic simulation (<0.8 nT ambient void)

Chamber 3: Matched Bifilar Sham Control

Magnetic Regime: Opposing windings; net B = 0

Experimental Purpose: Controls for ohmic heat, acoustic vibration & power-supply ripple

(b) Hardware Self-Magnetic Assessment Protocol (Power-State Mapping)

Subsystem Power-State Magnetic Characterization & Deconvolution

Subsystem 1: LED PAR Lighting Array (0–400 μmol/m²/s)

Power: 0 to 25 W (DC driver current 0.1–1.8 A)

Induced Field: ΔB_LED ≈ 2–15 μT; proportional to photon flux; balanced routing minimizes loop area

Subsystem 2: Fluidic & Nutrient Delivery Pumps / Valves

Power: 0 to 8 W (peristaltic/piezo motor pulses, 0.5–1.5 A peak)

Induced Field: ΔB_pump ≈ 8–35 μT; transient DC spikes during scheduled root module irrigation cycles

Subsystem 3: Thermal Control (Heaters & Thermoelectric Coolers)

Power: 0 to 30 W (resistive strips / TEC elements, 1.0–2.5 A continuous)

Induced Field: ΔB_thermal ≈ 12–48 μT; dominant DC contributor during cold lunar nights; bifilar loops required

Subsystem 4: Avionics, Telemetry & Sensor Power Bus

Power: 3 to 12 W (quiescent flight computer + RF comm transmission)

Induced Field: ΔB_bus ≈ 0.5–4 μT; static baseline shift + RF ripple; mapped via quiescent baseline calibration

$$\mathbf{B}_{\text{ambient}}(t) = \mathbf{B}_{\text{sensor}}(t) - \sum_{k} \Delta\mathbf{B}_k(P_k(t))$$: **Computational Self-Field Subtraction**

(c) Post-Flight Phenotypic Risk Disambiguation Framework

Operational Multi-Stressor Disambiguation Matrix for Space Plant Biology

Near-Null Magnetic Field (<0.8 nT)

Primary Biophysical Target:

Cryptochrome flavin radical pairs (FAD/TrpH•+);
mitochondrial [4Fe-4S] cluster electron transfer.

Diagnostic Phenotypic Markers:

- Emergence lag (Δt50 = +12 to 36 h); capacity -10 to -30%
- Repressed chlorophyll (HEMA1, POR) & anthocyanin (CHS)
- Mitochondrial ROS leak; induction of AOX1D & CAT3
- Normal nuclear DNA integrity (zero γ-H2AX foci)

Disambiguation & Rescue Strategy:

- Correlate with sample-level 3-axis canopy magnetometry
- Fully rescued by 45 μT vector re-application
- Completely insensitive to 1g centrifuge spinning

[Diagnostic: Rescued by 45 μT | Insensitive to 1g]

Lunar Partial Gravity (0.166 g)

Primary Biophysical Target:

Columella statolith sedimentation kinetics;
mechanosensitive ion channels (MSL); PIN polar auxin.

Diagnostic Phenotypic Markers:

- Germination capacity & greening rate are normal
- Directional root skewing, waving, & loss of g-angle
- Cell wall remodeling (pectin demethylesterification)
- Normal CRY signaling; localized apoplastic ROS (RBOH)

Disambiguation & Rescue Strategy:

- Fully rescued to Earth phenotype by 1g centrifuge
- Differential 1g vs μg controls in flight centrifuge
- Completely unaffected by magnetic vector modulation

[Diagnostic: Rescued by 1g | Insensitive to B-field]

Deep-Space Radiation (GCR / SPE)

Primary Biophysical Target:

Nuclear & organellar DNA double-strand breaks;
radiolytic hydroxyl free radical (OH•) overflow.

Diagnostic Phenotypic Markers:

- Severe DNA double-strand breaks & γ-H2AX foci
- ATM/ATR activation; chromosome aberrations & G2/M arrest
- Widespread systemic oxidative stress & lipid peroxidation
- Emergence fails only at acute lethal doses (>50 Gy)

Disambiguation & Rescue Strategy:

- Independent of magnetic field or gravity vector
- Strictly correlates with canopy radiation dosimeters
- Mitigated by shielding & chemical radioprotectants

[Diagnostic: Correlates with Dosimeter | DNA Breaks]